

Transforming logarithmic expressions

Three laws that turn products into sums, and the change of base that lets a calculator finish the job.

LESSON 59–60

2 × 40 MIN

ALGEBRA 10, §3.5

P2 · 2.2

LESSON CLOCK

12'

22'

22'

22'

20'

12'

The three laws	12 min
Expanding and condensing	22 min
Change of base	22 min
The three traps	22 min
Practice	20 min
Homework	12 min

LEARNING OBJECTIVES

- State and apply the three laws of logarithms.
- Expand and condense logarithmic expressions.
- Use the change-of-base formula.
- Recognise and avoid the three standard misapplications.

TEXTBOOK

National	pp. 237–248
Cambridge	Pure Mathematics 2 & 3, pp. 18–25
Workbook	P2 Exercise 2E

01

Explanation

The three laws

$\log_a(xy) = \log_a x + \log_a y$

$$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$$

$$\log_a(x^n) = n \cdot \log_a x$$

WHERE THEY COME FROM

A logarithm is an index, and $a^m \cdot a^n = a^{(m+n)}$. Multiplying the numbers **adds** the indices — so multiplying the arguments adds the logarithms. All three laws are index laws in disguise.

Before calculators, this is exactly how multiplication was done: look up two logarithms, add them, look the answer back up. The slide rule is a physical version of

the first law.

Expanding and condensing

EXPRESSION	EXPANDED
$\log(x^2y^3)$	$2 \log x + 3 \log y$
$\log\left(\frac{x}{y^2}\right)$	$\log x - 2 \log y$
$\log(x\sqrt{y})$	$\log x + \frac{1}{2} \log y$
$\log\left(\frac{x^3}{\sqrt{y}}\right)$	$3 \log x - \frac{1}{2} \log y$

Going the other way — **condensing** — is what solves equations. Coefficients become powers, sums become products, differences become quotients:

$$2 \log x + \log 3 - \log y = \log \frac{3x^2}{y}$$

Change of base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Any convenient base c may be used — usually 10 or e , because those are the two on a calculator:

$$\log_2 10 = \frac{\lg 10}{\lg 2} = \frac{1}{0.30103} \approx 3.32$$

Two useful special cases follow at once:

$$\log_a b = \frac{1}{\log_b a} \quad \log_{(a^n)} b = \frac{1}{n} \log_a b$$

The three traps

TRAP 1

$\log(x + y)$ is **not** $\log x + \log y$. There is no law for the logarithm of a sum. Test it: $\lg(10 + 10) = \lg 20 \approx 1.3$, but $\lg 10 + \lg 10 = 2$.

TRAP 2

$\frac{\log x}{\log y}$ is **not** $\log\left(\frac{x}{y}\right)$. The quotient law applies to the logarithm **of** a quotient, not to a quotient **of** logarithms.

TRAP 3

$(\log x)^2$ is not $\log(x^2)$. The first is a logarithm squared; the second is $2 \log x$. Watch where the bracket is.

These three account for almost every wrong answer in the topic. Read the expression, decide which law **could** apply, and if none does, leave it alone.

02 Worked examples

EXAMPLE 1 Expand $\log_2 \frac{8x^3}{\sqrt{y}}$.

$$\textcircled{1} \quad \log_2 8 + \log_2 x^3 - \log_2 \sqrt{y}$$

$$\textcircled{2} \quad = 3 + 3 \log_2 x - \frac{1}{2} \log_2 y$$

ANSWER

$$3 + 3 \log_2 x - \frac{1}{2} \log_2 y$$

EXAMPLE 2 Condense $3 \log x - 2 \log y + \frac{1}{2} \log z$.

$$\textcircled{1} \quad \log x^3 - \log y^2 + \log \sqrt{z}$$

$$\textcircled{2} \quad = \log \frac{x^3 \sqrt{z}}{y^2}$$

ANSWER

$$\log \frac{x^3 \sqrt{z}}{y^2}$$

EXAMPLE 3 Given $\lg 2 \approx 0.301$, find $\log_2 5$.

$$\textcircled{1} \quad \lg 5 = \lg \frac{10}{2} = 1 - 0.301 = 0.699$$

$$\textcircled{2} \quad \log_2 5 = \frac{\lg 5}{\lg 2} = \frac{0.699}{0.301}$$

$$\textcircled{3} \quad \approx 2.32$$

ANSWER

$$\approx 2.32$$

03 Interactive model

Test every "law" a learner invents on numbers before accepting it.

Is that a law?

$\log(xy)$ equals:

$$\log x \cdot \log y$$

$$\log x + \log y$$

$$\log x - \log y$$

$$(\log x)(\log y)$$

$\log(x + y)$ equals:

$$\log x + \log y$$

$$\log x \cdot \log y$$

none of these

$$\log x - \log y$$

$\log(x^5)$ equals:

$$5 \log x$$

$$(\log x)^5$$

$$\log 5 + \log x$$

$$\frac{\log x}{5}$$

$\log x$ $\log y$ equals:

$$\log\left(\frac{x}{y}\right)$$

$$\log x - \log y$$

$$\log_y x$$

$$\log x \cdot \log y$$

$\log_a b$ in base 10:

$$\frac{\lg a}{\lg b}$$

$$\frac{\lg b}{\lg a}$$

$$\lg b - \lg a$$

$$\lg(ab)$$

$(\log x)^2$ equals:

$$2 \log x$$

$$\log(x^2)$$

neither

$$\log x + \log x$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Law of logarithms	Logarifm qoidasi	Свойство логарифма
Product law	Ko'paytma qoidasi	Логарифм произведения
Quotient law	Bo'linma qoidasi	Логарифм частного
Power law	Daraja qoidasi	Логарифм степени
Change of base	Asosni almashtirish	Переход к новому основанию
Expand	Yoyish	Разложить
Condense	Bir logarifmga keltirish	Свернуть
Common logarithm	O'nli logarifm	Десятичный логарифм
Misapplication	Noto'g'ri qo'llash	Неверное применение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$\log_a x + \log_a y$ equals:

$3 \log x$ equals:

$\log_5 25 - \log_5 5$ equals:

Change of base $\log_a b$ is:

$$\frac{\log_c a}{\log_c b}$$

$$\frac{\log_c b}{\log_c a}$$

$$\log_c b \cdot \log_c a$$

$$\log_c(ab)$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\log_2 4 + \log_2 8$

ANSWER 5

2. $\log_3 27 - \log_3 3$

ANSWER 2

3. $\log_2 2^5$

ANSWER 5

4. Expand $\log(xy)$

ANSWER $\log x + \log y$

5. Expand $\log(x^3)$

ANSWER $3 \log x$

6. Condense $\log 2 + \log 5$

ANSWER $\log 10 = 1$

7. Condense $2 \log x$

ANSWER $\log x^2$

Medium · the standard question

7 PROBLEMS

1. Expand $\log_2 \frac{8x^3}{\sqrt{y}}$

ANSWER $3 + 3\log_2 x - \frac{1}{2}\log_2 y$

2. Condense $3 \log x - 2 \log y$

ANSWER $\log \frac{x^3}{y^2}$

3. $\lg 4 + \lg 25$

ANSWER 2

4. $\log_6 4 + \log_6 9$

ANSWER 2

5. $\log_2 10$ given $\lg 2 = 0.301$

ANSWER ≈ 3.32

6. $\log_2 5$ given $\lg 2 = 0.301$

ANSWER ≈ 2.32

7. $\log_9 27$

ANSWER 1.5

Hard · stretch and reason

7 PROBLEMS

1. Simplify $\log_2 3 \cdot \log_3 8$

ANSWER 3

2. Simplify $\frac{1}{\log_2 6} + \frac{1}{\log_3 6}$

ANSWER 1

3. Given $\lg 2 = a$ and $\lg 3 = b$, express $\lg 12$

ANSWER $2a + b$

4. Same: express $\lg 1.5$

ANSWER $b - a$

5. Simplify $\log_5 2 \cdot \log_4 25$

ANSWER 1

6. Solve $\log_2 x + \log_2(x - 2) = 3$

ANSWER $x = 4$

7. Show $\log_a b \cdot \log_b a = 1$

ANSWER Change of base twice

07 Homework

Homework — 6 tasks

Test any rule you are unsure of on actual numbers first.

- Expand $\log_3 \frac{9x^4}{y^2}$ and $\lg(x\sqrt{y})$.
- Condense $4 \log x + \frac{1}{2} \log y - 3 \log z$.
- Evaluate $\log_2 6 + \log_2 \frac{8}{3}$ and $\lg 50 + \lg 2$.

4. Given $\lg 2 = 0.301$ and $\lg 3 = 0.477$, find $\lg 6$, $\lg 1.5$ and $\log_3 2$.
5. Show that $\log_a b \cdot \log_b c \cdot \log_c a = 1$.
6. Give a numerical counterexample to each of the three traps.